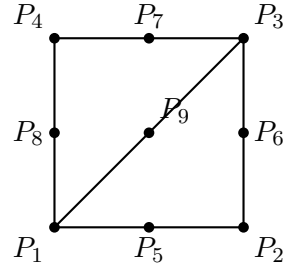


Poisson-Gleichung auf dem Einheitsquadrat

$$-\Delta u = 1 \quad \text{in } \Omega = [0, 1]^2$$

Schwache Form:

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} v \, dx$$



$$T_1 = \{(x, y) : y \leq x\}, \quad T_2 = \{(x, y) : y \geq x\}$$

Basisfunktionen und Gradienten

$$\varphi_1 = \begin{cases} (1-x)(1-2x), & y \leq x \\ (1-y)(1-2y), & y \geq x \end{cases} \quad \nabla \varphi_1 = \begin{cases} \begin{pmatrix} 4x-3 \\ 0 \end{pmatrix}, & y \leq x \\ \begin{pmatrix} 0 \\ 4y-3 \end{pmatrix}, & y \geq x \end{cases}$$

$$\varphi_2 = \begin{cases} (x-y)(2x-2y-1), & y \leq x \\ 0, & y \geq x \end{cases} \quad \nabla \varphi_2 = \begin{cases} \begin{pmatrix} 4x-4y-1 \\ -4x+4y+1 \end{pmatrix}, & y \leq x \\ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, & y \geq x \end{cases}$$

$$\varphi_3 = \begin{cases} y(2y-1), & y \leq x \\ x(2x-1), & y \geq x \end{cases} \quad \nabla \varphi_3 = \begin{cases} \begin{pmatrix} 0 \\ 4y-1 \end{pmatrix}, & y \leq x \\ \begin{pmatrix} 4x-1 \\ 0 \end{pmatrix}, & y \geq x \end{cases}$$

$$\varphi_4 = \begin{cases} 0, & y \leq x \\ (y-x)(2y-2x-1), & y \geq x \end{cases} \quad \nabla \varphi_4 = \begin{cases} \begin{pmatrix} 0 \\ 0 \end{pmatrix}, & y \leq x \\ \begin{pmatrix} -4y+4x+1 \\ 4y-4x-1 \end{pmatrix}, & y \geq x \end{cases}$$

$$\varphi_5 = \begin{cases} 4(1-x)(x-y), & y \leq x \\ 0, & y \geq x \end{cases} \quad \nabla \varphi_5 = \begin{cases} \begin{pmatrix} 4-8x+4y \\ -4+4x \end{pmatrix}, & y \leq x \\ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, & y \geq x \end{cases}$$

$$\varphi_6 = \begin{cases} 4(x-y)y, & y \leq x \\ 0, & y \geq x \end{cases} \quad \nabla \varphi_6 = \begin{cases} \begin{pmatrix} 4y \\ 4x-8y \end{pmatrix}, & y \leq x \\ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, & y \geq x \end{cases}$$

$$\varphi_7 = \begin{cases} 0, & y \leq x \\ 4x(y-x), & y \geq x \end{cases} \quad \nabla \varphi_7 = \begin{cases} \begin{pmatrix} 0 \\ 0 \end{pmatrix}, & y \leq x \\ \begin{pmatrix} 4y-8x \\ 4x \end{pmatrix}, & y \geq x \end{cases}$$

$$\varphi_8 = \begin{cases} 0, & y \leq x \\ 4(1-y)(y-x), & y \geq x \end{cases} \quad \nabla \varphi_8 = \begin{cases} \begin{pmatrix} 0 \\ 0 \end{pmatrix}, & y \leq x \\ \begin{pmatrix} -4+4y \\ 4-8y+4x \end{pmatrix}, & y \geq x \end{cases}$$

$$\varphi_9 = \begin{cases} 4(1-x)y, & y \leq x \\ 4(1-y)x, & y \geq x \end{cases} \quad \nabla \varphi_9 = \begin{cases} \begin{pmatrix} -4y \\ 4-4x \end{pmatrix}, & y \leq x \\ \begin{pmatrix} 4-4y \\ -4x \end{pmatrix}, & y \geq x \end{cases}$$

Finite-Element-System

Mit

$$A_{ij} = \int_{\Omega} \nabla \varphi_i \cdot \nabla \varphi_j \, dx$$

und

$$f_i = \int_{\Omega} \varphi_i \, dx$$

ergibt sich das lineare Gleichungssystem

$$Au = f$$

mit

$$A = \begin{pmatrix} 1 & \frac{1}{6} & 0 & \frac{1}{6} & -\frac{2}{3} & 0 & 0 & -\frac{2}{3} & 0 \\ \frac{1}{6} & 1 & \frac{1}{6} & 0 & -\frac{2}{3} & -\frac{2}{3} & 0 & 0 & 0 \\ 0 & \frac{1}{6} & 1 & \frac{1}{6} & 0 & -\frac{2}{3} & -\frac{2}{3} & 0 & 0 \\ \frac{1}{6} & 0 & \frac{1}{6} & 1 & 0 & 0 & -\frac{2}{3} & -\frac{2}{3} & 0 \\ -\frac{2}{3} & -\frac{2}{3} & 0 & 0 & \frac{8}{3} & 0 & 0 & 0 & -\frac{4}{3} \\ 0 & -\frac{2}{3} & -\frac{2}{3} & 0 & 0 & \frac{8}{3} & 0 & 0 & -\frac{4}{3} \\ 0 & 0 & -\frac{2}{3} & -\frac{2}{3} & 0 & 0 & \frac{8}{3} & 0 & -\frac{4}{3} \\ -\frac{2}{3} & 0 & 0 & -\frac{2}{3} & 0 & 0 & 0 & \frac{8}{3} & -\frac{4}{3} \\ 0 & 0 & 0 & 0 & -\frac{4}{3} & -\frac{4}{3} & -\frac{4}{3} & -\frac{4}{3} & \frac{16}{3} \end{pmatrix}$$

und

$$f = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \frac{1}{6} \\ \frac{1}{6} \\ \frac{1}{6} \\ \frac{1}{6} \\ \frac{1}{3} \end{pmatrix}$$

Also:

$$A \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \\ u_7 \\ u_8 \\ u_9 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \frac{1}{6} \\ \frac{1}{6} \\ \frac{1}{6} \\ \frac{1}{6} \\ \frac{1}{3} \end{pmatrix}$$

Lokale P_2 -Basisfunktionen

Für ein Dreieck mit baryzentrischen Koordinaten

$$\lambda_1, \lambda_2, \lambda_3$$

lauten die quadratischen lokalen Basisfunktionen:

$$\phi_1 = \lambda_1(2\lambda_1 - 1)$$

$$\phi_2 = \lambda_2(2\lambda_2 - 1)$$

$$\phi_3 = \lambda_3(2\lambda_3 - 1)$$

$$\phi_4 = 4\lambda_1\lambda_2$$

$$\phi_5 = 4\lambda_2\lambda_3$$

$$\phi_6 = 4\lambda_1\lambda_3$$

Für das betrachtete Gebiet gilt:

$$T_1 = \{(x, y) : y \leq x\}$$

mit

$$\lambda_1 = 1 - x, \quad \lambda_2 = x - y, \quad \lambda_3 = y$$

sowie

$$T_2 = \{(x, y) : y \geq x\}$$

mit

$$\mu_1 = 1 - y, \quad \mu_2 = y - x, \quad \mu_3 = x$$

Daraus ergeben sich die globalen Basisfunktionen:

$$\varphi_1 = \begin{cases} \lambda_1(2\lambda_1 - 1), & y \leq x \\ \mu_1(2\mu_1 - 1), & y \geq x \end{cases}$$

$$\varphi_2 = \begin{cases} \lambda_2(2\lambda_2 - 1), & y \leq x \\ 0, & y \geq x \end{cases}$$

$$\varphi_3 = \begin{cases} \lambda_3(2\lambda_3 - 1), & y \leq x \\ \mu_3(2\mu_3 - 1), & y \geq x \end{cases}$$

$$\varphi_4 = \begin{cases} 0, & y \leq x \\ \mu_2(2\mu_2 - 1), & y \geq x \end{cases}$$

$$\varphi_5 = \begin{cases} 4\lambda_1\lambda_2, & y \leq x \\ 0, & y \geq x \end{cases}$$

$$\varphi_6 = \begin{cases} 4\lambda_2\lambda_3, & y \leq x \\ 0, & y \geq x \end{cases}$$

$$\varphi_7 = \begin{cases} 0, & y \leq x \\ 4\mu_2\mu_3, & y \geq x \end{cases}$$

$$\varphi_8 = \begin{cases} 0, & y \leq x \\ 4\mu_1\mu_2, & y \geq x \end{cases}$$

$$\varphi_9 = \begin{cases} 4\lambda_1\lambda_3, & y \leq x \\ 4\mu_1\mu_3, & y \geq x \end{cases}$$